

Attention Cycles

Joel Flynn, Karthik Sastry

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Some context for the paper

“What we have is widely dispersed knowledge, which cannot be concentrated in one mind.”

- F. Hayek, *The use of knowledge in society*, 1945.

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- In many models, welfare losses up to second order:

$$\begin{aligned}\mathbb{E}_t \mathcal{U}_t &\approx -\frac{(1-\sigma)}{2} \underbrace{\mathbb{E}_t[(Y_t - Y_t^*)^2]}_{\approx \text{Volatility of Output gap}} - \frac{1}{2} \underbrace{\mathbb{E}_t \left[\int_I (y_{it} - Y_t)^2 di \right]}_{\approx \text{Dispersion in Prices}} \\ &\approx -\frac{(1-\sigma)}{2} \underbrace{\mathbb{E}_t \left[\int_I (y_{it} - Y_t^*)^2 di \right]}_{\approx \text{Volatility+Dispersion}} - \frac{\sigma}{2} \mathbb{E}_t \left[\int_I (y_{it} - Y_t)^2 di \right]\end{aligned}$$

with $0 < \sigma < 1$.

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with $0 < \sigma < 1$.

- With *exogenous* information frictions:

$$y_{it} = Y_t^* + \sigma \mathbb{E}_{it}(Y_t - Y_t^*) \quad (1)$$

with $\int_I \mathbb{E}_{it}(Y_t - Y_t^*) di = 0$ (“on average correct”).

- Key for equation (1) is GE + dispersed information.
- Three models of information frictions:
 1. Quantities are chosen before observing prices, with flex prices.
 2. Prices are chosen before observing quantities, with flex quantities.
 3. Extensive vs intensive margins / inventories and rationing.

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- What happens when information is endogenous?

1. *Aggregate* value of information is cyclical

$$y_{it} = Y_t^* + \sigma_t(\mathbf{Y}_t) \mathbb{E}_{it}(Y_t - Y_t^*) \quad (2)$$

2. With heterogeneity z_{it} , cyclical dispersion in *individual* values of information

$$y_{it} = y_{it}^* + \sigma_{it}(\mathbf{Y}_t, \mathbf{z}_{it}) \mathbb{E}_{it}(Y_t - Y_t^*) \quad (3)$$

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$$y_{it} = y_{it}^* + \sigma_{it}(\mathbf{Y}_t, \mathbf{z}_{it}) \mathbb{E}_{it}(Y_t - Y_t^*) \quad (3)$$

- Theoretical model with heterogeneity and endogenous information:

$$y_{it} = y^*(z_{it}) + \sigma_{it}(Y_t, z_{it})\varepsilon_{it}, \quad \varepsilon_{it} \sim N(0, 1) \quad (4)$$

$\Rightarrow \sigma_{it}(Y_t, z_{it})$ is *procyclical* in the model.

- “Empirical evidence” about *procyclical*ity of variance of misoptimization

$$\mathbb{E}_t \left[\int_I (y_{it} - y_{it}^*)^2 di \right]$$

conditional on the mechanism of the model.

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Mechanism of the model

- Firm maximizes expected discounted profits subject to a “information collection cost:”

$$\max_{y_{it}, p_{it}=D(y_{it})} \mathbb{E}_{it} \left[\pi(y_{it}, z_{it}) \underbrace{Y_t^{-\gamma}}_{\text{SDF}} - \underbrace{\mathcal{C}(y_{it}, z_{it})}_{\text{Cost in utils}} \right]$$

by choosing the distribution of $y_{it} \sim \underbrace{f(\cdot|z_{it})}_{\text{Stochastic rule}} = N(y_{it}^*, \sigma_{it}^2)$.

- When Y_t is low, the SDF is high, and the “price of mistakes” is high:
 $\Rightarrow$ countercyclical incentives to make *precise* decisions.

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Details of the model

- Nice form for y_{it} follows from likelihood-separable assumption on

$$\mathbb{E}_{it} \mathcal{C}(y, z_{it}) := \lambda_i \underbrace{\int_{\mathcal{Z}} \int_{\mathcal{Y}} \left\{ f(y|z_{it}) \log f(y|z_{it}) \right\} \cdot dy \cdot dF(z_{it})}_{\text{Relative entropy}}$$

- Which implies the usual form for $f(y_{it}|z_{it})$

$$\Rightarrow f(y_{it}|z_{it}) = \frac{\exp(\lambda_i^{-1} \pi(\mathbf{y}_{it}, \mathbf{z}_{it}) \mathbf{Y}_t^{-\gamma})}{\int_{\mathcal{Y}} \exp(\lambda_i^{-1} \pi(y', \mathbf{z}_{it}) \mathbf{Y}_t^{-\gamma}) dy'}$$

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- Define the full-information “frictionless” benchmark

$$(y_{it}^*, p_{it}^*) := \arg \max_{y, p=D(y)} \left[\pi(y, z_{it}) Y_t^{-\gamma} \right]$$

- Up to a second-order approximation around $(y_{it}^*, p_{it}^*, \mathbf{Y}_t, \mathbf{P}_t)$

$$\pi(y_{it}, z_{it}) Y_t^{-\gamma} = \pi(y_{it}^*, z_{it}) \mathbf{Y}_t^{-\gamma} + \frac{1}{2} \pi_{yy}(y_{it}^*, z_{it}) \mathbf{Y}_t^{-\gamma} (y_{it} - y_{it}^*)^2$$

(Q: Does the point $(y_{it}^*, p_{it}^*, Y_t, P_t)$ exist? Are supporting beliefs “rational?”)

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- Plug profits into $f(y_{it}|z_{it})$

$$f(y_{it}|z_{it}) \propto \exp\left(-\frac{(y_{it} - y_{it}^*)^2}{2\lambda_i |\pi_{yy}(y_{it}^*, z_{it})|^{-1} Y_t^\gamma}\right)$$

- Hence the specification from equation (4)

$$y_{it} = y^*(z_{it}) + \underbrace{\sqrt{\frac{\lambda_i}{|\pi_{yy}(y^*(z_{it}), z_{it})| Y_t^{-\gamma}}}}_{=: \sigma_{it}(Y_t, z_{it})} \varepsilon_{it}, \quad \varepsilon_{it} \sim N(0, 1)$$

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Rest of the model

- The rest is standard RBC:
 - Linear production technology with labor:

$$y_{it} = \theta_{it} \ell_{it}$$

- Representative consumer:

$$\mathcal{U}_t := \mathbb{E}_t \left\{ \frac{C_t^{1-\gamma}}{1-\gamma} - \nu(L_t) \right\}$$

- CES aggregator:

$$Y_t = \left(\int_0^1 y_{it}^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}}$$

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Rest of the presentation

- Empirical evidence informative about $\mathbb{E}_t(Y_t - Y_t^*)$:

Claim (1): Macro Attention is *countercyclical*.

- Empirical evidence informative about $\mathbb{E}_t[\int_I (y_{it} - y_{it}^*)^2 di]$:

Claim (2): Misoptimization dispersion is *procyclical*.

Claim (3): Mistakes are more costly in downturns.

- Discussion.

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Claim (1): Macro Attention is countercyclical

- Data: Quarterly 10-Q and annual 10-K SEC reports, US public firms.
- Period: 1995 - 2018.
- Sample: Around 5000 documents per quarter.
- Method: Text analysis to detect frequency of macroeconomic language.

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- Example from General Motors, 2009:

“The deteriorating economic and market conditions that have driven the drop of vehicle sales, including declines in real estate and equity values, rising unemployment, tightened credit markets, depressed consumer confidence and weak housing markets, may not improve significantly during 2010 and continue past 2010 and could deteriorate further.”

Claim (1): Macro Attention is countercyclical

- Details of method: *term frequency inverse document frequency*

$$\text{tf-idf}(w)_{it} := \text{tf}(w)_{it} \cdot \log \left(\frac{1}{\text{df}(w)} \right)$$

- $\text{tf}(w)$: *term frequency* for word w in document of firm i at time t .
 - $\text{df}(w)$: *document frequency* for word w among all regulatory filings.
 - $\text{tf-idf}(w)_{it}$ is high when word w is frequent in the document.
- Macro Attention is measured as

$$MA_{it} := \sum_{w \in \mathcal{W}^{\text{macro}}} \text{tf-idf}(w)_{it}$$

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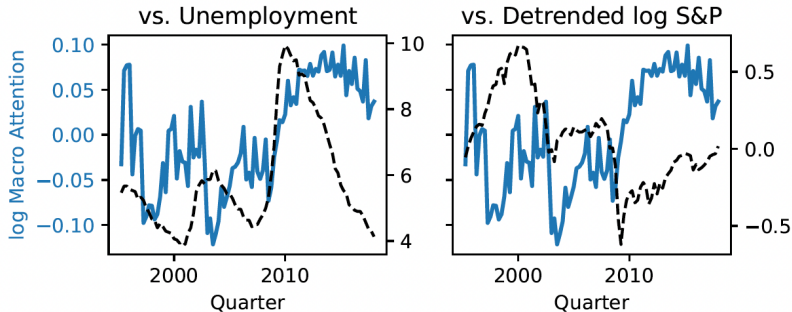
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Notes: The blue line, measured by the left axis of each plot, is Macro Attention in log, seasonally-adjusted units. The black dashed lines, measured by the right axis of each plot, are respectively the U.S. unemployment rate and linearly detrended S&P 500 price.

News

Industries

Words

Claim (2): Misoptimization dispersion is procyclical

- Data: Compustat Annual.
- Period: 1986 - 2016.
- Sample: About 2200 firms per year.
- Method: Estimation of mistakes as residuals.

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- Details of method: Assumptions

- Cobb-Douglas production technology to estimate "Solow residual":

$$\log y_{it} = \underbrace{\log \theta_{it}}_{\text{TFP}} + \alpha_s^{\ell} \underbrace{\log \ell_{it}}_{\text{Labor}} + \alpha_s^m \underbrace{\log m_{it}}_{\text{Materials}} + \alpha_{it}^k \underbrace{\log k_{it}}_{\text{Capital}}$$

- Policy function for labor:

$$\log \ell_{it} = \underbrace{\eta_i}_{\text{Firm}} + \underbrace{\chi_{s,t}}_{\text{Industry} \times \text{Time}} + \beta \log \theta_{it} + \underbrace{m_{it}}_{\text{Misoptimization}}$$

- Persistent misoptimization:

$$m_{it} = \rho m_{it-1} + \left(\sqrt{1 - \rho^2} \right) \underbrace{u_{it}}_{\text{Innovations}}$$

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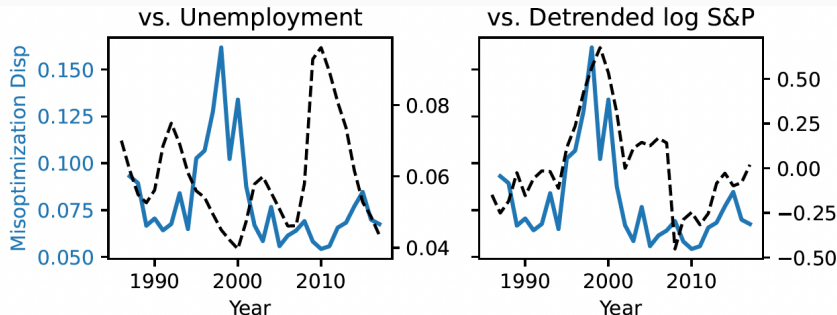
- The measure of misoptimization dispersion is:

$$\mathbb{E}_t \left[\int_I (y_{it} - y_{it}^*)^2 di \right] \propto \frac{\sum_i s_{it} \cdot \hat{u}_{it}^2}{\sum_i s_{it}}$$

where s_{it} is the predicted sale based on fundamentals:

$$\log \text{Sales}_{it} = \beta \log \theta_{it} + \eta_i + \chi_{s,t} + \nu_{it}$$

Claim (2): Misoptimization dispersion is procyclical



Notes: The blue line, measured by the left axis of each plot, is Misoptimization Dispersion as defined by Equation 28. The black dashed lines, measured by the right axis of each plot, are respectively the unemployment rate and detrended S&P 500 price.

Adjustment cost

Financial frictions

Other inputs

Other robustness

Survey evidence

Claim (3): Mistakes are more costly in downturns

- Markets punish (squared?) mistakes when returns are low:

$$\Delta \log P_{it} = \beta \hat{u}_{it}^2 + \phi \cdot (\hat{u}_{it}^2 \times \Delta \log P_t) + \chi_{st} + \Gamma' X_{it} + \nu_{it}$$

- The channel goes only through lower profits:

$$\pi_{it} = \beta_{\pi} \hat{u}_{it}^2 + \phi_{\pi} \cdot (\hat{u}_{it}^2 \times \Delta \log P_t) + \chi_{st} + \gamma_i + \nu_{it}$$

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Table 1: Markets Punish Misoptimization Harder in Low-Return States

	(1)	(2)	(3)	(4)
	Outcome: $\Delta \log P_{it}$			
$\hat{u}_{it}^2$	-0.268 (0.025)	-0.262 (0.023)	-0.097 (0.034)	-0.087 (0.033)
$\hat{u}_{it}^2 \times \Delta \log P_t$	0.376 (0.123)	0.376 (0.124)	0.443 (0.171)	0.431 (0.167)
Sector x Time FE	✓	✓	✓	✓
Firm FE			✓	✓
TFP Control		✓		✓
N	41,578	41,578	41,206	41,206
R^2	0.239	0.261	0.385	0.403

Notes: Standard errors are double-clustered at the year and firm-ID level.

Claim (3): Mistakes are more costly in downturns

Table 2: Misoptimization, Profits, and Pricing

	(1) Outcome: π_{it}	(2)	(3)	(4)
		Outcome: $\Delta \log P_{it}$		
$\hat{u}_{it}^2$	-0.114 (0.020)	-0.021 (0.032)		
$\hat{u}_{it}^2 \times \Delta \log P_t$	0.112 (0.089)			
π_{it}		0.400 (0.028)	0.421 (0.034)	0.690 (0.305)
$\pi_{it} \times \Delta \log P_t$			-0.303 (0.166)	-1.642 (0.632)
Firm FE	✓	✓	✓	✓
Sector x Time FE	✓	✓	✓	✓
N	50,966	40,879	40,879	40,879
R^2	0.663	0.402	0.402	
First-stage F				17.80

Notes: Standard errors are double-clustered at the year and firm-ID level.

Linking claim (1) and (2)

Table 3: Macro-Attentive Firms Make Smaller Misoptimizations

	(1)	(2)	(3)	(4)
		Outcome: $\hat{u}_{it}^2$		
log MacroAttention _{it}	-0.0081 (0.0028)	-0.0052 (0.0029)	-0.0058 (0.0044)	-0.0056 (0.0038)
Sector x Time FE	✓	✓	✓	✓
Firm FE			✓	✓
TFP, Price Controls		✓		✓
<i>N</i>	28,279	24,392	27,875	23,930
<i>R</i> ²	0.053	0.067	0.383	0.384

Notes: Standard errors are double-clustered at the year and firm level.

Discussion and conclusions

- Empirical evidence: what are they picking up?
- Information acquisition and uncertainty:
 1. Benhabib, Liu, Wang (JET, 2016): countercyclical uncertainty.
 2. Mäkinen and Ohi (JET, 2015): countercyclical uncertainty.
 3. Bloom et al. (ECTA, 2018): countercyclical dispersion in firm-level productivity.
- But, theoretical point likely to hold in general settings. How much does it matter for welfare in practice?
- Possible connection with static calculations of Hopenhayn and Venkateswaran (QJE, '16), David and Venkateswaran (AER, '19).

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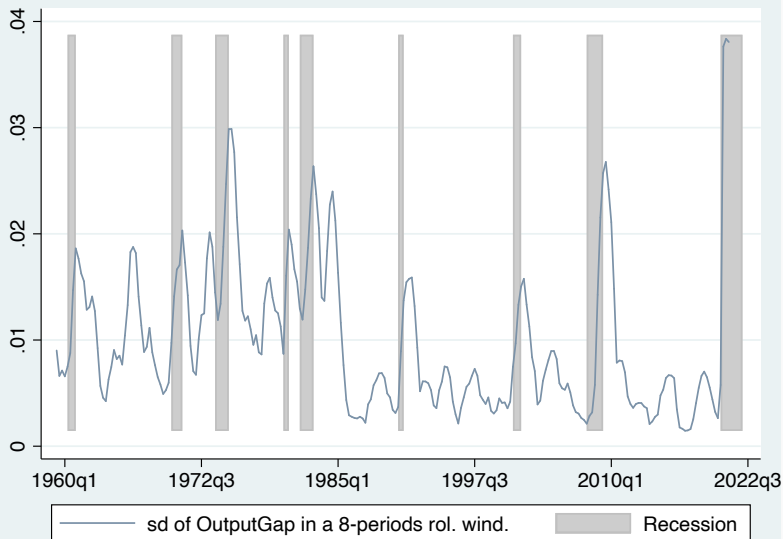
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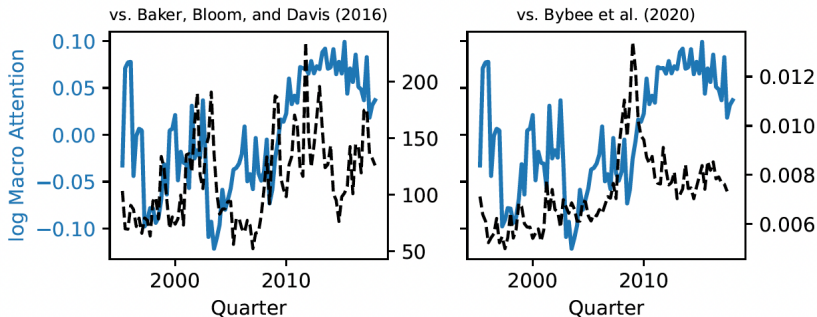
Time series of CBO Output Gap volatility



Appendix

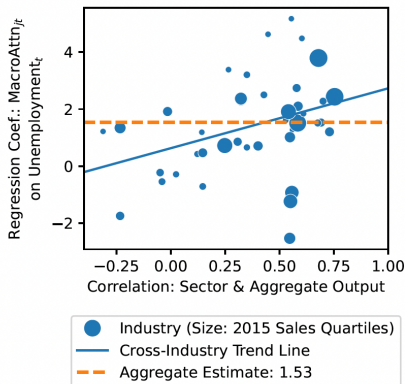
Comparison with news indexes

Figure A3: Relationship of Macro Attention to News Indices



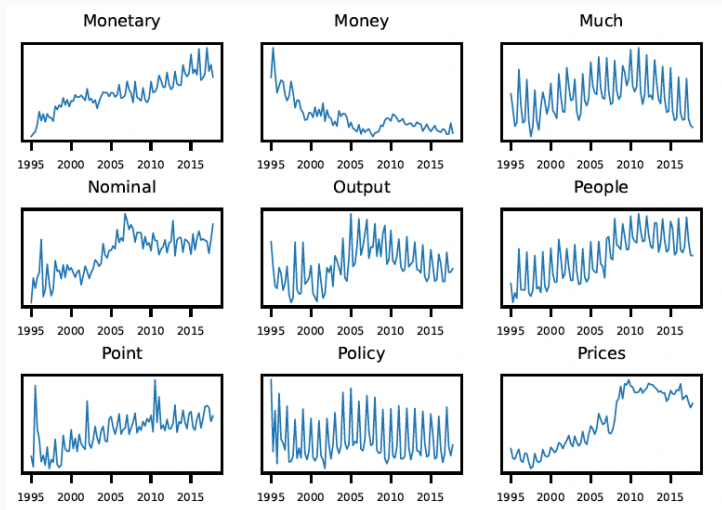
Notes: The left panel is the news index of [Baker et al. \(2016\)](#). The right panel is the sum of five macroeconomic topics from the [Bybee et al. \(2021\)](#) Structure of News dataset, described in Section 2.3 of the main text.

Figure A4: Industry-Specific Cyclicity of Macro Attention



Notes: The horizontal axis is the correlation of sectoral and aggregate nominal GDP, calculated as described in Section 2.3 of the main text. The vertical axis is the regression coefficient of log sectoral macro attention, net of quarterly fixed effects, on the US unemployment rate. The dashed orange line is the estimate of the same using aggregate Macro Attention. The dots are sized based on quartiles of total sales in Compustat in 2015. The blue solid line is a cross-industry linear regression line.

Data construction: example



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- Industry at 3-digit NAICS for information and manufacturing, 2 for rest.
- Materials: $\text{cogs} + \text{xsga} - \text{dp} - \text{wage bill}$.
- Capital: Ottonello Winberry (2020).
Plants + property + equipment + net investment.
- 15-20% of US employment, 60-80% of US output.

1. Input elasticities as cost shares in each sector.
2. Productivity as Solow residual.
3. Estimate labor policy to get m_{it} .
4. Estimate dynamics of m_{it} to get ρ .
5. Re-estimate policy + dynamics calibrating ρ to get u_{it} .

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- Policy function for labor including lag:

$$\log \ell_{it} = \underbrace{\eta_i}_{\text{Firm}} + \underbrace{\chi_{s,t}}_{\text{Industry} \times \text{Time}} + \beta \log \theta_{it} + \tau \log \ell_{it-1} + \underbrace{m_{it}}_{\text{Misoptimization}}$$

- Coeff on unemployment: -.43 (.19), lower than benchmark -.84 (.34).

- Policy function for labor including leverage:

$$\log \ell_{it} = \underbrace{\eta_i}_{\text{Firm}} + \underbrace{\chi_{s,t}}_{\text{Industry} \times \text{Time}} + \beta \log \theta_{it} + \tau L_{it} + \phi \cdot (L_{it} \times \log \theta_{it}) + \underbrace{m_{it}}_{\text{Misoptimization}}$$

- Coeff on unemployment: -.84 (.34), same as benchmark -.84 (.34).

- Total cost expenditure (i.e. materials + labor).
- Investment rate (i.e. log growth rates of capital stock).

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1. Sector-specific policy functions.
2. TFP $\times$ year fixed effect.
3. Non-linearities in TFP.
4. Time-varying output elasticities.
5. Pre-sample for policy, post-sample for mistakes.
6. Productivity using Olley and Pakes (1996).
7. Linear and quadratic time trend control.
8. Only manufacturing sector.

- Survey evidence on firms in New Zealand by Coibon et al. (2018).
- Firms report to pay more attention to news about macroeconomy if there is a negative shock.
- Higher sensitivity of firm profits to their choices in downturns.

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